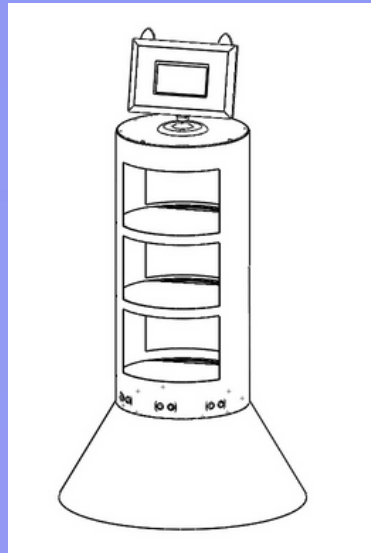


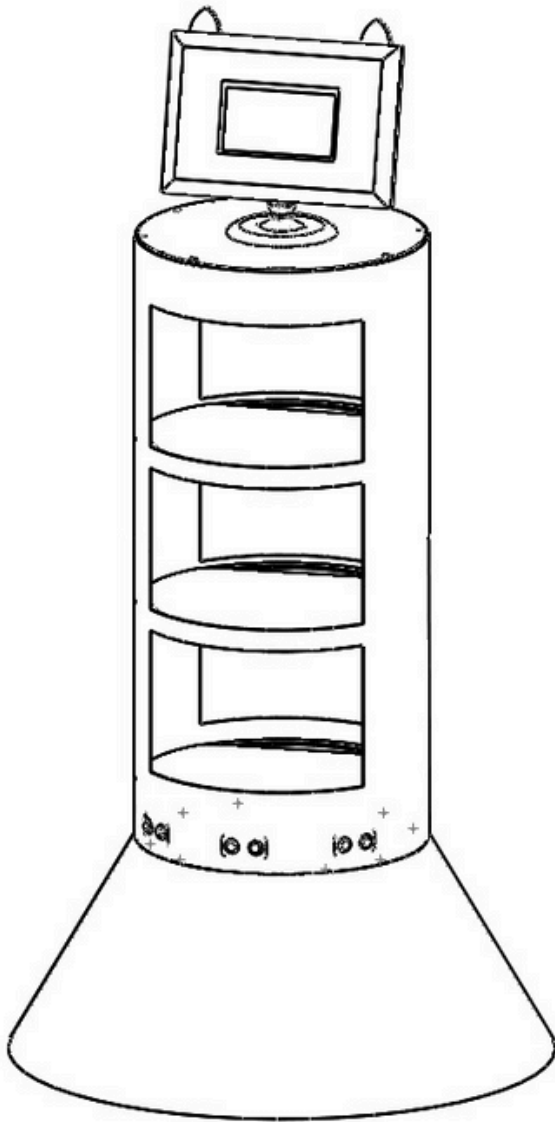
RESTAURANT SERVICE ROBOT

USER MANUAL



PREPARED BY

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INTRODUCTION

Welcome to the user manual for the Restaurant Robot, an intelligent and autonomous service unit designed to assist in food delivery within restaurants. This robot minimizes human effort, optimizes table service, and ensures a contactless, efficient dining experience.

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- Autonomous navigation with metal-strip tracking
 - Automated food delivery and return-to-kitchen function
 - Safe customer interaction and obstacle avoidance
 - Long battery life with smart charging interface

KEY FEATURES

Parameter	Specification
Operating Voltage	24V
Charging Time	30min
Battery Lifetime	30min
Cruise Speed	0.1ms ⁻¹
Number of Trays	3
Tray Load	5kg
Material	Stainless Steal
Battery Capacity	6200mAh
Weight	50kg
Dimensions(LxWxH)	80cmx80cmx145cm

WORKING MECHANISM

Food Delivery Mode – Step-by-Step Process

In delivery mode, the robot is capable of serving multiple tables in a single run. Food for different customers is placed on separate trays, and the robot plans the most efficient path to deliver them sequentially. Once deliveries are complete, it automatically returns to the food pick-up point.

1. Dish Loading

The waiter places the food items on the trays, assigning different trays to different orders if necessary.

2. Mode and Table Selection

On the touchscreen, the waiter selects Delivery Mode. Then, they tap the tray icon and assign a table number for each one. The robot registers and maps out the route accordingly.

3. Task Start

Once all table numbers are entered, the waiter presses Start. The robot begins navigating autonomously.

4. Navigation

It follows a predefined path using metal strips on the floor:

- **Single metal strip:** Normal navigation path
- **Double metal strip:** Indicates junctions to detect turns and table locations

5. Arrival & Positioning

Upon reaching the assigned table, the robot:

- Rotates to face the customer directly
- Announces arrival with a voice prompt
- Displays a message on the screen inviting the customer to retrieve their food

6. Food Pickup by Customer

Customers take their food. They may tap the screen to confirm completion, or the robot resumes automatically after a brief pause.

7. Return to Kitchen

Once all deliveries are completed, the robot returns to the starting position and waits for the next task.

After the customer touches the “Return” button on the screen, the robot navigates back to the kitchen area autonomously.

POWER & CHARGING

Power ON, Power OFF, Pause, and Start Instructions

- **Before Each Power-On:** Position the robot directly under the visual mark for accurate initialization.
- **Power-On:** Hold the power button for 0.5 seconds. A blue light will appear at the base.
- **Startup:** The touchscreen will enter working mode to indicate successful boot.
- **Power-Off:** Press and hold the power button for 3 seconds to shut down. For a forced shutdown, hold for 10 seconds (only if the robot is unresponsive).

Charging Instructions

- **To begin charging:**
 - Connect the robot's charging port to the provided charger.
 - Ensure a secure connection. A charging prompt will appear on the robot's screen when connected successfully.

Power Requirements

1. Keep battery level above 10% during operation to ensure optimal battery health.
2. When the battery falls below 10%, the robot enters low-power mode and should be charged promptly.
3. If the battery level drops below 2%, the robot will stop all functions to protect the battery and must be recharged before use.

- **Battery Type:** Lithium-Ion
- **Charging Port:** DC Input (use only original adapter)
- **Battery Status:** Displayed via screen and front LED
- **Low Power Warning:** Robot returns to charge when battery < 20%

POWER & CHARGING

Power and Charging Requirements

1. Charge the robot when battery drops below 20% to avoid long-term battery degradation.
2. Always ensure the battery is properly installed before connecting the charger.
3. Do not remove the detachable battery while charging.
4. Never remove the battery while the robot is powered on.
5. Once fully charged, disconnect the power to avoid overcharging.
6. If not used for an extended period, switch off and lock the robot to protect the battery.
7. Only use manufacturer-provided battery and charging accessories.
8. Confirm the voltage input matches the rating on the charger label.
9. Only use the original charger for the detachable battery; battery charge level is shown via indicator lights.
10. Ensure the robot is powered off before swapping batteries.
11. Using incorrect voltage may damage the charger; always verify compatibility.
12. Handle charging cables gently—avoid pulling or twisting them.
13. Assign a responsible person to oversee charging. Do not leave the robot charging unattended.
14. Do not charge near flammable or explosive items.
15. Always store and charge in a dry, cool area (below 40°C). Keep water away.
16. Prevent the charger from any physical damage or strong impacts.
17. Replace the charger immediately if damaged or showing irregular current.
18. If the robot shows a charging-related error, disconnect the power immediately.

**Note: Avoid leaving the robot charging unattended overnight.
Ensure a dry and ventilated charging area.**

SAFETY INSTRUCTIONS

1. Never exceed the maximum load: 10kg per tray (standard) or 5kg per tray (when covered).
2. Always press the "Finish" button only after customers have collected their food to start the next task.
3. Do not pick up or place food while the robot is moving. Use the screen to pause the robot first. Note: pause duration is 20 seconds in cruise mode and 10 seconds in other modes.
4. Do not press forcefully or tap aggressively on the touchscreen.
5. Never pull the robot manually during its operation.
6. Avoid pushing the robot backward while it is powered on.
7. If the robot deviates from its route due to obstacles, pause it and gently guide it back to the correct path before resuming.
8. In case of a collision, pause or cancel the task. Do not try to force or block the robot.
9. Unless using a special tray, avoid delivering soups. If used, ensure containers are sealed to prevent burns or spillage.
10. Keep a safe distance when the robot is transporting hot items to avoid collisions or burns.
11. The top camera is essential for navigation. Do not obstruct it with any covering.
12. Avoid hitting, shaking, or interacting roughly with the robot, as this may damage its components.
13. Children must interact under adult supervision.
14. Do not place or charge the robot near flammable or explosive materials.
15. Ensure the robot is stored and charged in a dry, temperature-controlled environment (below 40°C). Prevent water or moisture from entering the machine or charger.
16. Protect the charger from collisions or external damage.
17. Replace the charger if it shows signs of damage or abnormal charging current.
18. Immediately disconnect the charging power source if a robot error or alarm is triggered.
19. Pause the robot before pushing or relocating it.
20. Never pour liquids into any part of the robot.
21. Do not place objects, children, or pets on the robot at any time.
22. Although equipped with obstacle avoidance, never intentionally block the robot during fast movement to avoid accidents.
23. Be aware of the robot's blind-spot detection limitations and avoid interfering within its low-ground detection zone.

PRODUCT MAINTENANCE

Trays, Drive Wheels, and Auxiliary Wheels

- Clean the trays regularly using a soft cotton cloth.
- Inspect and clean the drive and auxiliary wheels weekly.
- If debris is stuck in the wheels, gently lift the robot to remove it safely.

Sensor Maintenance

- Clean the top-positioning sensor and 3D obstacle-avoidance sensors weekly.
- If contaminated, clean immediately using lens-safe tissue or cloth.
- Obstructed sensors may cause abnormal movement or positioning errors.

Robot Body Maintenance

- Wipe the robot's surface with a dry or lightly damp cotton cloth.
- Never lift, bump, climb on, or apply force to the robot.
- Do not stack any objects on the robot.
- Do not disassemble any part of the robot unless instructed by a certified technician.

Handling the Robot

- During transport, use proper equipment (e.g., forklifts) that meets GB/T 4857.23-2012 standards.
- If manually moved, lift the robot from both sides (as marked) using two people.
- Maintain balance and keep the robot upright. Never lift by holding the tray.

Sign Maintenance

- Do not hang any items (balloons, banners, etc.) directly under or near the ceiling signs.
- Avoid damaging or repositioning the signs during ceiling maintenance.

CONTACT & SUPPORT

For technical support or warranty service, contact:

- **Email:** msboralu@gmail.com
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Thank you for choosing the Restaurant Robot. For questions, support, or training, don't hesitate to reach out!